Day 36 Practice Problems: Deep Learning Foundations & Perceptrons (Hinglish)

Yeh practice suite teen production-grade engineering challenges provide karti hai taaki aap Day 36 mein seekhe gaye artificial neuron mathematics, non-linear decision boundaries, PyTorch tensor forward pass, aur deep learning frameworks ko master kar sakein.

Challenge 1: Diagnostic Neuron ka Manual Calculation aur PyTorch Proof

Background & Context

Aap ek hospital ke liye cardiovascular risk prediction model bana rahe ho. Ek single artificial neuron patient ke 4 normalized health parameters process karta hai:

1. Normalized Age ($x_1 = 0.60$)
2. Normalized Blood Pressure ($x_2 = 0.75$)
3. Normalized Cholesterol ($x_3 = 0.80$)
4. Normalized Max Heart Rate ($x_4 = 0.50$)

Neuron ke trained weights aur bias diye gaye hain:

$$\mathbf{w} = \begin{bmatrix} 0.40 \\ 0.70 \\ 0.50 \\ -0.60 \end{bmatrix}, \quad b = -0.15$$

Tasks & Sawal

1. **Manual Linear Sum (z)**:
   - Weighted sum ka formula likhein aur exact step-by-step value calculate karein:

$$z = \sum_{i=1}^{4} w_i x_i + b$$

2. **Sigmoid Activation Se Probability ($\hat{y}$)**:

- Value z ko Sigmoid function mein pass karke predicted probability nikaalein:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

- Agar threshold 0.50 hai, toh kya patient high risk par classified hoga?

3. **PyTorch Code Implementation**:
   - PyTorch ke `nn.Linear(in_features=4, out_features=1)` ka use karke yahi neuron banayein.
   - Is layer ke `.weight` aur `.bias` ko manual values se initialize karein.
   - Input tensor `[0.60, 0.75, 0.80, 0.50]` pass karein aur verify karein ki PyTorch ka output aapke manual calculation se exact match karta hai.

Challenge 2: Non-Linearity aur XOR Problem (Perceptron ki Limit)

Background & Context

1969 mein Marvin Minsky aur Seymour Papert ne prove kiya tha ki ek akela Perceptron **XOR gate** ko kabhi solve nahi kar sakta kyunki XOR non-linear hota hai aur ek single seedhi line se divide nahi ho sakta.

XOR Truth Table:

x_1	x_2	**Target Output** y	**Point** (x_1, x_2)
0	0	0	$(0, 0)$
0	1	1	$(0, 1)$
1	0	1	$(1, 0)$
1	1	0	$(1, 1)$

Tasks & Sawal

1. **Mathematical Contradiction Proof**:
   - Charon points ke liye equations banakar mathematically prove karo ki aisi koi weights w_1, w_2 aur bias b exist nahi karte jo charon equations ko ek sath satisfy kar sakein.
2. **Multi-Layer Architecture (MLP)**:

- Ek 2-layer neural network (2 inputs, 2 hidden neurons, 1 output neuron) design karo jo XOR problem ko solve kare.
- Har neuron ke weights aur bias clearly define karo.

Challenge 3: Industrial Framework Sizing & Architecture

Background & Context

Ek e-commerce company 10 million industrial product images ke liye automated quality inspection pipeline build kar rahi hai ($1024 \times 1024 \times 3$ RGB images).

Tasks & Sawal

1. **ML vs Deep Learning Selection**:
   - 3 solid engineering reasons dijiye ki traditional machine learning (SIFT + SVM) is scale par kyun fail ho jayegi aur Deep Learning kyun zaroori hai.
2. **PyTorch vs TensorFlow vs Keras**:
   - Agar R&D team ko naye custom attention layers banane hain aur deployment team ko edge devices par model chalana hai, toh kaunsa framework chunna chahiye aur kyun?
3. **GPU VRAM Calculation**:
   - Batch size 32 ke ek single forward pass batch ka raw memory footprint calculate karein (`float32` precision). Training ke waqt raw data se 3x-4x zyada VRAM kyun chahiye hoti hai?

Solutions Reference

Step-by-step verified calculations aur PyTorch code ke liye Solutions_HINGLISH.md check karein.